

day-3

Date: 01.11.25

Agenda

① EDA (Exploratory data analysis)

↳ Analyzing a dataset to understand

- ↳ main characteristics
- ↳ identify patterns
- ↳ discover relationship between variables
- ↳ detect anomalies using statistics and visualization.

② Feature engineering

- ↳ column selection
- ↳ transformation
- ↳ missing value handle

③ model training

↳ by doing hyper parameter tuning.

modular coding

- ① Data ingestion
- ② Data validation (Data schema)
- ③ Data transformation
- ④ model training
- ⑤ model evaluation
- ⑥ model registry
- ⑦ model deployment

Industry documentation:-

- ① HLD → High level Documentation
- ② LLD → Low level documentation.

Today procedure :-

- ① Understand the problem.

→ we need to understand our data type and based on this data we need to solve our ~~the~~ problem.

This is visa prediction related and we ~~solve~~ predict, ~~if a person~~ will a person visa approve or not?

⑪ Data collection:-

we collect the data from kaggle.
where has 25480 row and 12
columns and it is csv file.

⑬ Reasons for Performing EDA:-

- we have 12 column, where some columns are numerical and some columns are categorical. so first of all, i need to separate them.
- convert the categorical data into numerical data.
- lets check the data columns data which column for vs to model training and predict data.
 - if a column most of the row/data is unique, then we don't need it.
 - else, we need it
- Univariate Analysis
 - it means explore one by one feature / column.

→ eda perform in numerical columns
→ eda perform categorical columns.

→ In numerical feature we are use `sns.kdeplot()`.

→ In categorical feature we are use `sns.countplot()`, (chi2-test)

all data are (mainly case status)
not stable, that means ~~not~~ imbalance.

→ In numeric and categorical ^{feature} data
we check is it continuous or
discrete.

~~Now check target column and~~

Now check Input feature and
target column relation.

So, what we exactly do in EDA?

- ① Understand structure
- ② Detect missing or incorrect data
- ③ Understand distribution
- ④ Find relationship
- ⑤ prepare for modeling

So for this EDA, step is —

- ① Load and inspect the data
- ② Handle missing values
- ③ check data balance (Target variable)
 - if target classes are highly imbalanced
 - ① Oversampling (SMOTE)
 - ② Undersampling
 - ③ Weighted loss
- ④ Univariate analysis (one variable at a time)
 - numerical
 - categorical
- ⑤ Bivariate analysis (Relation between two variables)
 - numerical vs numerical (cont. vs dis.)
 - numerical vs categorical
 - categorical vs categorical

vi) Correlation analysis
↳ for feature selection

vii) Outlier Detection
↳ apply transformation.

viii) Feature engineering ideas
↳ create new features (e.g., ratios, interactions)
↳ convert categorical variables (encoding)
↳ Normalize or scale numeric
↳ combine or bin similar categories.